HVAC ENGINEERING VALIDATION REPORT

Automated ISHRAE / ECBC / NBC Compliance Audit System

STATUS: NON-COMPLIANCE DETECTED

1. Project Profile Context

Location Context Zone:	Delhi
Space Usage Profiling:	Office
Evaluated Net Area:	500.0 Sq. Meters
Target Personnel Count:	2 Occupants

2. Thermodynamic Sizing Matrix

Calculated Scientific Load Requirement:	12.11 TR
User Specified Design Capacity:	45.0 TR
Structural Capacity Deviation:	271.59%

3. Code & Standard Infractions

- CRITICAL OVERSIZING: Designed capacity (45.0 TR) is 271.59% higher than the calculated peak scientific requirement (12.11 TR). This will cause compressor short-cycling, poor humidity control, and high electricity bills.

- ENVELOPE NON-COMPLIANCE: Wall U-Value is 0.65 W/m^{2}K, which exceeds the max limits set by India's Energy Conservation Building Code (ECBC) max limit of 0.4 W/m^{2}K.

- EFFICIENCY DEFICIT: Equipment Coefficient of Performance (COP) is 3.0. Modern high-efficiency systems should meet or exceed a COP of 4.5 under AHRI standard conditions.

4. Value Engineering & Energy Optimization

- SUGGESTION: Downsize equipment capacity closer to 12.11 TR. Implement a Multi-Stage Compressor or a Variable Frequency Drive (VFD) chiller pack. Estimated annual operational electricity savings: ~Rs. 710,424.0.

- SUGGESTION: Add 50mm of Rockwool insulation or extruded polystyrene (XPS) boards to external walls to lower the U-value. This directly downsizes structural cooling infrastructure requirements.